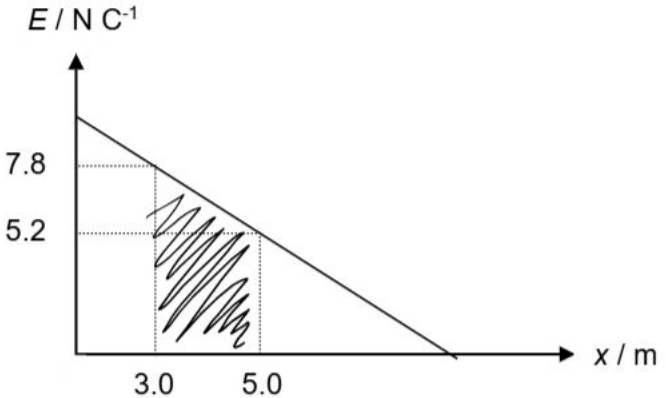
18 The graph below shows how the electric field strength E varies with displacement x from a point A. What is the change in potential for an electron if it is moved from a point 3.0 m away from A to another point 5.0 m away from A?



A −13 V **B** −1.3 V **C** 1.3 V **D** 13 V

Ans: A

Electric field strength, $E = -\frac{dV}{dx}$

Therefore, change in potential can be calculated from the area under this graph,

$$dV = \text{Area of Trapezium} = \frac{1}{2}(7.8+5.2)(5.0-3.0) = 13\,\text{V}$$

From the graph, magnitude of the electric field strength is positive, which implies that the direction of the electric field strength is in the same direction as the displacement (electric field lines point from high to low potential). This implies that potential decreases as the displacement increases.

Therefore, required change in potential is -13 V. The negative sign shows that the potential decreases.